

ion and electron conducting (MIEC) garnet material. The team has presented the improved rate capability of the lithium (Li) metal anode in this study. This level of performance was attainable due to a novel 3D design and a distinctive architectural strategy. This technique sets the stage for a fresh approach to designing the forthcoming era of high-energy rechargeable batteries. The successful showcasing of high-rate, dendrite-free lithium metal within 3D MIEC structures is expected to inspire the development of viable solid-state batteries using a 'Li-free' anode methodology.

Qibbot: The quickest human-operated combat robot

Shandong, China-based Qibot Robot company, claims they have created the world's swiftest boxing robot. At over 1.9-metre (six feet) in height, this single-handed robot can



Qibbot in action during its match with Ai-guided bot (Credit: <https://interestingengineering.com>)

be remotely controlled and boasts a response delay of merely 12 milliseconds. The team presented the ambitious task of constructing a large-scale robot boasting low latency. To

overcome the bot's mechanical and controller challenges and enhance speed and performance, they innovatively integrated a 'feed-forward' controller with the standard feedback controller in the control algorithm. This new controller was designed to predict system lags, acting proactively to counteract the latency induced by mechanical systems and the virtual reality tools employed by human operators. This new approach was crafted to predict and proactively counteract latency from mechanical systems and virtual reality tools.

A robotic system for weed control and crop monitoring

Researchers at the University of Bonn have developed a robotic system called BonnBot-I that may aid farmers in more efficiently managing weeds and monitoring crops. The robot employs localisation sensors, including GPS and odometry, enabling it to navigate fields, locate, classify, count plants, and manage weeds using integrated tools. The system seamlessly integrates with the Robot Operating System. The team curated a new dataset for training algorithms to identify and count corn, a challenging crop for computer

vision detection. The researchers leveraged crop monitoring techniques, integrating GNSS and wheel odometry to boost precision in plant location and classification.

This enhancement reduced tracking accuracy error from

8.3% to 3.5%, assessed on a new publicly accessible corn dataset. They also introduced a unique setup of weeding tools on linear actuators, tested in simulations.

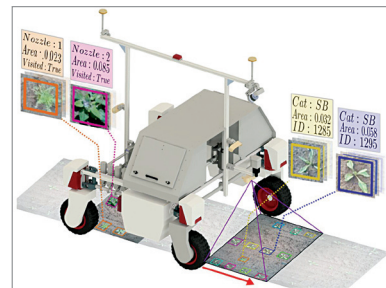


Figure illustrating the BonnBot-I platform (Credit: Ahmadi, Halstead & McCool)

Wearable ultrasound scanner: Early breast cancer detection

MIT researchers have created wearable ultrasound gear to improve breast cancer survival rates by detecting early tumours. The team aims 98% survival via frequent screening. They made a compact ultrasound scanner using innovative material. A flexible, 3D-printed patch with magnets attaches it to a bra for skin contact, allowing versatile breast imaging at six positions with adjustable angles.



In hopes of improving the breast cancer survival rate, MIT researchers designed a wearable ultrasound device that could allow patients to detect tumors when they are still in early stages (Credit: Canan Dagdeviren)

Tested on a 71-year-old with small cysts, the device detects early tumours. Linked to standard ultrasound, it's becoming smartphone-like. Reusable patch for home use, aiding high-risk or screening-limited cases, matches traditional ultrasound's resolution, views 8cm deep, improving comfort and reliability in breast cancer detection. Researchers seek AI-driven analysis for evolving diagnostics, surpassing radiologist image comparisons.

AI model enhances trust and reduces bias

Researchers at the University of Waterloo have designed an artificial intelligence (AI) model that mitigates bias, amplifies trustworthiness and precision in decision-making and machine learning (ML)-driven knowledge organisation.